DR. SARAH CONNOR

CONTACT

Remote
+1 800 555 0100
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CORE SKILLS

- Python
- Rust
- Go
- Distributed Systems
- Real-Time Control Systems
- Machine Learning / AI
- Systems Architecture

EDUCATION

Ph.D., Robotics & Autonomous Systems
Carnegie Mellon University

M.S., Computer Science
Stanford University

CERTIFICATIONS

Certified Kubernetes Administrator
AWS Certified Machine Learning – Specialty

Specialist in Autonomous Intelligence Systems

12 years of experience · Research & Engineering Leadership

PROFILE

Research-driven engineering leader with 12 years of experience building autonomous intelligence systems, from low-level real-time control in Rust and Go to large-scale machine learning infrastructure in Python. Track record of taking systems from research prototype to production deployment in safety-critical environments. Skilled at leading cross-functional teams and translating cutting-edge research into robust, scalable software.

EXPERIENCE

Principal Engineer, Autonomous Systems — Cyberdyne Robotics 2019 – Present

- Architected a real-time perception and decision-making pipeline in Rust and Go, deployed across a fleet of 500+ autonomous units with 99.99% uptime.
- Led a team of 10 engineers and researchers building ML-driven navigation systems in Python, cutting incident rate by 55%.
- Designed the distributed systems backbone coordinating fleet-wide decision-making, processing 50M+ sensor events per day.
- Set technical strategy and roadmap for autonomous intelligence R&D, presenting to executive leadership and external partners.

Senior Software Engineer, Robotics — Skynet Dynamics 2015 – 2019

- Built core control-system software in Go for autonomous mobile platforms operating in unstructured environments.
- Developed Python-based simulation and training infrastructure that accelerated model iteration cycles by 3x.
- Collaborated with hardware teams to integrate sensor fusion algorithms into real-time embedded control loops.

Research Engineer — Advanced Robotics Lab, CMU 2012 – 2015

- Conducted doctoral research on autonomous decision-making under uncertainty, published in top-tier robotics venues.
- Prototyped early control algorithms in Python and C++ later adopted into industry autonomous systems.

SELECTED PUBLICATIONS

Connor, S. et al. “Real-Time Decision Architectures for Autonomous Agents.” Journal of Robotics Research.

Connor, S. et al. “Distributed Sensor Fusion for Fleet-Scale Autonomy.” IEEE Conference on Robotics & Automation.